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65/-

Department of Applied Chemistry

SGS Institute of Technology and Science, Indore

I Mid Term, April-2022

Subject Code- CH10506 (CHEMISTRY)

Time: 60 Minutes

Note: All Questions are compulsory.

MM: 15

S. No.	Questions 2.8 kg, 3.01 kg	Marks	CO	BL	PI
Q.1	Calculate the amount of lime (92% pure) and soda (98% pure) required for treatment 40,000 litres of water, whose analysis is as follows: $Ca(HCO_3)_2$=41.5 ppm, $Mg(HCO_3)_2$=58.55 ppm, $MgSO_4$=30.00ppm, $CaSO_4$=34.0ppm, NaCl=20.0ppm, KCl=20.0ppm, $CaCl_2$=27.75 ppm	5	2	2,3	1.2.1
Q.2	Draw the structures of ion (cation and anion) exchange resins. And Giving reaction of each, explain the function of the following in watertreatment (a) Calgon (b) Sodium aluminate	2.5+2.5	2	1,2	1.3.1
Q.3	Find out the number of fundamental vibrations of the following- Chloroform, Benzene, Ammonia, Carbon-di-oxide, Formaldehyde. OR Write down the note on absorption and intensity shifts of UV-Visible spectroscopy.	5	4	1,2, 3	4.1.2

Department of Applied Chemistry

S.G.S.I.T.S, Indore

Test-2, May-2022

Subject Code-CH10506 (CHEMISTRY)

MM:15

Time: 60 Minutes

Note: All questions are compulsory

S. No.	Questions	Marks	CO	BL
Q.1	100 ml water sample required 8ml of N/50 H_2SO_4 for neutralization to phenolphthalein end-point. Another 18 ml of the same acid was needed for further titration to methyl orange end-point. Determine the type and amount of alkalinity.	5	2	1,3
Q.2	a. An oil of unknown viscosity-index has a Saybolt universal viscosity of 58 seconds at 210° F and of 580 seconds at 100° F. The Pennsylvanian oil has Saybolt viscosity of 58 seconds at 210°F and 430 seconds at 100°F. The Gulf oil has a Saybolt universal viscosity of 58 seconds at 210°F and 780 seconds at 100° F. Calculate the viscosity-index of unknown oil.	2.5	3	1,3
	b. Write short notes on: passivity and galvanic corrosion.	2.5	3	1
Q.3	Write the function of the following regarding atomic absorption spectroscopy: 1. Nebulizer 2. Hollow cathode lamp 3. Monochromator 4. Atomizer 5. Photomultiplier tube	5	4	1,2
	(Or) What is chromatography? Write the principal and significance of column chromatography?	5	4	1

July-Dec.-2019-I

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B. Tech 1st Year Sections-H, I and J

Mid Semester Exam I, 2019-20

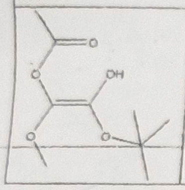
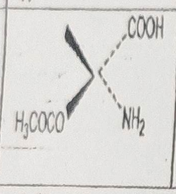
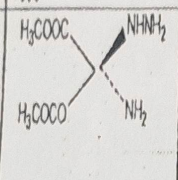
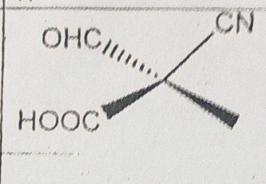
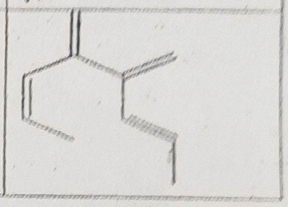
CHEMISTRY CH10506

Time: 1 Hr. MM: 30

Q.1. Subparts "a" and "b" are compulsory, Answer any Three (3) from the remaining subparts

a. Draw the structure of (2R,3R,4R,5R)-2,3,4,5,6-Pentahydroxyhexanal and (S)-2-Amino-3-phenylpropanoic acid in Wedge/Dash Presentation. 2

b. Assign absolute configuration to the following compounds 5

i.	ii	iii	iv	v
				

c. Give hybridization of the central atom: ClF_3, $Pt(CN)_4]^{2-}$ 1

d. Suggest name of a polymer 1

i) which prefers anionic mechanism of polymerization.

ii) which prefers cationic mechanism of polymerization.

e. Which of the following salts cause permanent hardness of water 1

i) $BaCl_2$ ii) $CaSO_4$ iii) $CsCl$ iv) $BeCO_3$ v) $Fe_2(SO_4)_3$ vi) $Mg(HCO_3)_2$ vii) $Al_2(SO_4)_3$

f. State two differences between addition and condensation polymerization. 1

g. Write down the structures of two initiators for free radical polymerization. 1

h. Write down two units of hardness. 1

i. Explain briefly why He_2 does not exist using M.O. diagram. 1

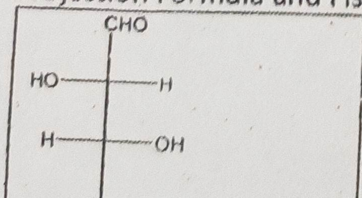
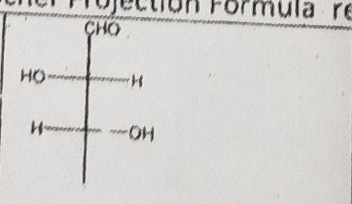
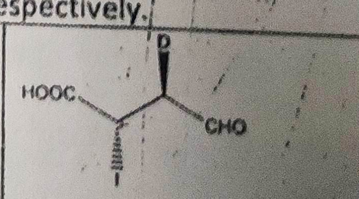
j. Write down the synthesis of Nylon 6,6. 1

k. Mention two drawbacks of Zeolite method of water softening. 1

l. Briefly explain Ion exchange method of water softening. 1

Q. 2. Subpart "a" is compulsory, Attempt any Three (3) from the remaining subparts. 4x3=12

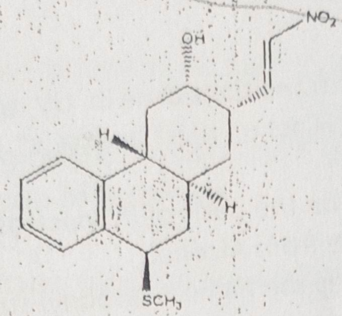
a. Convert the following compounds into Newmann Projection Formula, Sawhorse Projection Formula and Fischer Projection Formula respectively.

b. Explain why O_2 is paramagnetic.

c. What is the significance of Glass Transition Temperature.

d. Identify the chiral centers and assign absolute configuration of those centers.



e. Explain the concept of biodegradable polymer with examples.
f. Explain Soda-Lime method mentioning equations.
g. Classify polymer (with example) according to tacticity, origin and structure.
h. What do you mean by Caustic Embrittlement? Explain briefly.
i. How many grams of $MgCl_2$ dissolved in water, will give hardness of 76 ppm.
j. Explain coordinate bonding with example.

Q.3. Attempt any two of the following questions. 2x4=8

a. A sample of water has been found to contain the following salts $Ca(HCO_3)_2$ = 0.0105 gm, $Mg(HCO_3)_2$ = 0.0125 gm, $CaCl_2$ = 0.0082 gm, $MgSO_4$= 0.0026 ~~mg~~ gm, $CaSO_4$= 0.0075 ~~mg~~ gm
Calculate the temporary, permanent and total hardness.

b. When 50 mL of standard hard water consisting of 1 mg/ mL of pure $CaCO_3$ was titrated against EDTA it consumed 20 mL of it, but when a water sample of 50 mL consumed 25 mL of the same EDTA solution. 50 mL of water sample after boiling consumed 18 mL of the same EDTA. Calculate temporary, permanent and total hardness of the water sample.

c. Calculate the strength of the solution in molality prepared from 23.52 gm of NaCl in 5 kg of water.

d. 40,000 L of water to be softened. On analysis of water it was found to contain following impurities.
$Ca(HCO_3)_2$= 0.03 gm, $Mg(HCO_3)_2$= 0.09 gm, $MgSO_4$= 0.035 gm, $MgCl_2$= 0.16 gm, NaCl= 0.12 gm, $CaSO_4$=0.025 gm. Calculate the amount of lime and soda required. Lime is 89% pure and soda 92% pure. Ca= 40, Mg= 24

e. A water sample (500 mL) has a hardness equivalent for 50 mL of 0.1 N $CaSO_4$. What is the hardness in ppm?

f. A solution of hydrogen peroxide is 15.2% by mass. What is the molarity of the solution? Assume that the solution has a density of 1.01 g/mL.

SHRI G. S. INSTITUTE OF TECHNOLOGY AND SCIENCE
I MID TERM TEST JANUARY 2019
B.E IST YEAR CH-10506 CHEMISTRY

TIME: 1HOUR MM:15

Q.1 Discuss EDTA method for determination of hardness in water under following heads: 5
(a) Structure of EDTA-metal complex, (b) Structure of Indicator, (c) Principle, (d) Role of Buffer and (e) Calculation of Hardness.

OR

Enlisting special features, discuss mechanism of Free Radical and Coordination polymerization 5

Q.2 Define Polymers. Giving examples, discuss classification of polymers on the basis of: 5
(a) Tacticity (b) Thermal Behavior (c) Polymeric Structure (d) Structure of Backbone

Q.3 Calculate the quantities of Lime and Soda required for softening of 1000 litres of water using 13.9 ppm $FeSO_4.7H_2O$ as a coagulant. The results of the analysis of raw water and softened water are as follows: 5
Analysis of raw water: Ca^{++}=16.0ppm; Mg^{++}=28.8ppm; HCO_3^- =14.64ppm; CO_2= 20ppm. Analysis of treated water: CO_3^{2-}= 0.60ppm; OH^-=0.34ppm

Shri G.S. Institute of Technology and Science, Indore

Department of Applied Chemistry and Chemical Technology

B.E. I Year CH: 10506 (Chemistry)

I Mid Term Test October 2018

Time: 1 hr | MM: 15

Attempt any three questions

Q1. Calculate the quantities of lime and soda required for cold softening of 125,000 L of water with the following analysis, using 10 ppm of sodium aluminate as coagulant: (5)
Analysis of raw water: Ca^{2+} = 95 ppm, Mg^{2+} = 36 ppm, CO_2 = 66 ppm, HCO_3^- = 264 ppm, H^+ = 2 ppm
Analysis of treated water: CO_3^{2-} = 45 ppm, OH^- = 34 ppm.

Q2. Explain the difference between zeolite and ion exchange methods for softening of water. (5)

Q3. Write the preparation, properties and uses of any two of the following: (5)
- (i) Nitrile rubber
- (ii) Kevlar
- (iii) Bakelite

Q4.(a) How can the acid number of an oil be determined? Discuss its significance. (3)

(b) An oil sample under test has a Saybolt Universal viscosity same as that of standard Gulf oil (low viscosity standard) and Pennsylvanian oil (high viscosity index standard) at 210°F. Their Saybolt Universal viscosities at 100°F are 61, 758 and 420 s respectively. Calculate viscosity index of the sample oil. (2)

Q5.(a) Discuss the terms bathochromic and hypsochromic shifts. (2)

(b) A solution of thickness 3 cm transmits 30% incident light. Calculate the concentration of the solution, given extinction coefficient $\varepsilon = 4000\ dm^3\ mol^{-1}\ cm^{-1}$. (3)

Q6. Write short notes on any two: (5)
- (i) Aniline point
- (ii) Electrochemical mechanism of corrosion
- (iii) Classification of polymer on basis of effect of temperature
- (iv) Mechanism of free radical polymerization

SHRI G. S. INSTITUTE OF TECHNOLOGY AND SCIENCE

I MID TERM TEST JANUARY 2017

B.E IST YEAR CH10502 CHEMISTRY

TIME: 1HOUR MM:15

NOTE: Attempt any three questions out of seven.

Q.1 Discuss EDTA method of determination of hardness under following heads: (a) Structure of EDTA-metal complex (b) Structure of indicator(c) Theory (d) Roll of buffer (e) Calculation — 5

Q.2 Discuss any five boiler troubles in detail under following points: — 5
- (a) Causes
- (b) Disadvantages
- (c) Removal methods

Q.3 Giving examples discuss classification of polymers on the basis of: (a) Tacticity (b) Temperature effect (c) Types of polymerization. — 5

Q.4 Define hardness and discuss it's types. A sample of water on analysis was found to contain the following impurities: — 5

$Ca(HCO_3)_2$ = 4 mg/L

$Mg(HCO_3)_2$ = 6 mg/L

$CaSO_4$ = 8 mg/L

$MgSO_4$ = 10 mg/L

Calculate the temporary, permanent and total hardness of water in ppm, °Fr, °Cl.

Q.5 Enlisting functions of lubricants, discuss the mechanism of Thick film and Thin film lubrication with suitable examples. — 2+3

Q.6 With the help of neat and labeled diagram of bomb calorimeter, explain How the calorific value of a solid fuel can be determined? Discuss various corrections employed in it. — 5

Q.7 A) What is Bio-degradation of polymers? Discuss various steps involved in Bio-degradation of polymers. — 3+2

B) Discuss structure, properties and uses of KEVLAR.

SHRI G.S. INSTITUTE OF TECHNOLOGY & SCIENCE, INDORE

CLASS TEST I, SEPTEMBER 2016

B.E. I YEAR

PAPER: CHEMISTRY (CH10502)

Max. Marks: 15 Max. Time: 01 Hr.

Note: Attempt any three questions. All questions carry equal marks.

Q.1 a) Define (i) Polymer (ii) Degree of polymerization — 2 + 3

b) Discuss any three types of classifications of polymers.

Q.2 Define various types of calorific value. Give labeled diagram of Bomb calorimeter. Describe briefly various types of corrections employed for Calorific value determined by Bomb calorimeter. — 5

Q.3 a) Discuss hardness of water under following heads: Definition, Types and Reasons. — 2.5 +

b) A water sample on chemical analysis gave the following concentration of different salts: — 2.5

$Ca(HCO_3)_2 = 16.2$ mg/L; $Mg(HCO_3)_2 = 0.0073$ g/L;

$MgCl_2 = 9.5$ mg/L; $CaSO_4 = 13.6$ mg/L.

Calculate the temporary, permanent and total hardness in mg/L, ppm, °Cl, °French.

Q.4 Write short notes: — 3 + 2

(a) Proximate analysis of coal (b) Pulverization of coal

Q.5 Discuss following boiler troubles:

(a) Scales and sludges — 1.5 +

(b) Boiler corrosion — 1.5 +

(c) Caustic embrittlement — 2

Q.6 Discuss structure, property and uses of the followings: — 2 + 2 + 1

(a) Kevlar (b) Polyester (c) Polythene

Jan. 2016

SHRI G.S. INSTITUTE OF TECHNOLOGY & SCIENCE
CLASS TEST I
B.E. I YEAR
PAPER: CHEMISTRY (CH10502)

Date: 21st JAN. 2016

Max. Marks: 20 | Max. Time: 01 Hr.

Note: Attempt any two questions out of the followings

Q.1 Define polymers. Describe various classifications of polymers (at least four). Discuss methods of polymerization by taking suitable examples. **10**

Q.2 Write short notes on (ANY TWO): **10**

a) Pulverization of coal.

b) Proximate analysis of coal.

c) Low temperature and high temperature carbonization of coal.

Q.3 What do you mean by hardness of water? Discuss various types of water hardness. Calculate the temporary hardness, permanent hardness and total hardness (in mg/L, ppm, ^{0}Cl, ^{0}Fr) in a given water sample which on analysis gave the following concentration of hardness producing salts: **10**

$Ca(HCO_3)_2$ = 16.2 mg/L; $Mg(HCO_3)_2$ = 0.0073 g/L;
$MgCl_2$ = 9.5 mg/L; $CaSO_4$ = 13.6 mg/L.

Q.4 Give the criteria for selection of good fuel: coal or metallurgical coke. Describe the bomb calorimeter method for determination of calorific value with labeled diagram. **10**

Q.5 Discuss following boiler troubles: **10**

a) Scales and sludges. b) Boiler corrosion.

c) Caustic embrittlement. d) Priming and foaming.

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SHRI SG INSTITUTE OF TECHNOLOGY AND SCIENCE

II Mid Term Test 2011 CH-1058 (Section-E)

MM-20

Time-1 Hour

Attempt any four questions. All carry equal marks.

Q.1 Giving a suitable example explain Electrochemical Theory Of Corrosion.

Q.2 Discuss the Zeolite Process for the removal of hardness of water.

Q.3 What are the essential requirements of an explosive?

Q.4 Explain how corrosion can be minimized by the application of protective coatings.

Q.5 Write short note on- (ANY TWO)

a) Aniline Point of an oil

b) Cutting Fluids

c) Acid Number of an Oil